

E-LOCK: Automated Lockout/Tagout (LOTO) System

Integrating Wireless Relays and Biometric Authentication for Secure Electrical Isolation

1. Project Overview

E-LOCK is a modern, IoT-enabled Lockout/Tagout (LOTO) safety system designed for high-risk industrial environments. Traditional LOTO systems rely on manual management of physical padlocks and tags, which are highly susceptible to human error, procedural delays, and miscommunications .

E-LOCK addresses these systemic vulnerabilities by digitizing the authorization process and automating physical electrical isolation . By integrating **ESP32 microcontrollers**, **biometric scanners**, and **shunt trip circuit breakers**, the system ensures that machinery is completely de-energized and can only be restored after verified physical presence and digital clearance.

2. Key Features

- **Biometric Authorization:** Industrial equipment is protected by non-transferable fingerprint verification using the AS608 scanner, ensuring only registered personnel can toggle power.
- **Deterministic Safety Logic:** The system utilizes rigid hardware-level logic rather than unpredictable software layers to guarantee a 100% predictable safety environment during maintenance.
- **Transient-Filtered Failsafe:** To prevent "nuisance trips" caused by motor spin-down back-EMF, a **ZMPT101B Voltage Sensor** and a **Time Delay Relay (TDR)** introduce a hardware-level grace period before evaluating a fault.
- **Mechanical Override (Shunt Trip):** If the voltage sensor detects live power after the specified time delay (indicating a welded relay or bypassed contactor), it automatically fires the Shunt Trip to mechanically force the Main Circuit Breaker into the OFF position.

- **High-Speed Wireless Response:** Uses **ESP-NOW** for instant hardware-to-hardware communication between nodes, bypassing the need for a local Wi-Fi router for critical safety commands.
 - **Live Dashboard & Auditing:** A centralized web interface tracks real-time isolation status and maintains an automated, tamper-proof audit trail in a **PostgreSQL** database.
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3. Technology Stack

Web Dashboard (FSD Architecture)

- **Framework:** Next.js (App Router)
- **UI/UX:** Shadcn UI & TailwindCSS
- **Database:** PostgreSQL (Cloud/Local)
- **ORM:** Drizzle ORM
- **Communication:** WebSockets / HTTP API

Firmware & Embedded

- **Environment:** PlatformIO (VS Code)
- **Microcontrollers:** Dual ESP32 Setup (Gateway & Field Controller)
- **Protocols:** ESP-NOW (Node-to-Node) and Wi-Fi (Node-to-Server)
- **Language:** C++

Hardware Components

- **Safety:** Main Circuit Breaker with Shunt Trip Accessory, Time Delay Relay (TDR).
 - **Sensors:** AS608 Fingerprint Scanner, ZMPT101B AC Voltage Sensor.
 - **Control:** 5V Control Relays, Pull Box, Battery backup .
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4. Project Structure (Feature-Sliced Design)

Plaintext

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E-LOCK/

```
|— web/                # Next.js Web Dashboard
|   |— src/
|   |   |— app/        # App Router (Pages, Layouts, API Routes)
|   |   |   |— auth/    # Secure worker/admin authentication
|   |   |   |— admin/    # Personnel registration & system manual control
|   |   |   |— dashboard/ # Real-time LOTO monitoring & feedback
|   |   |   |— layout.tsx # Root layout with Sidebar/Header
|   |   |— components/  # Shared UI (Shadcn primitives)
|   |   |— config/      # Env vars & Drizzle/Postgres connection
|   |   |— db/          # PostgreSQL Schema & Drizzle Migrations
|   |   |— features/    # Domain-specific business logic
|   |   |   |— loto-status/ # Logic for machine-side feedback & failsafes
|   |   |   |— user-mgmt/  # Biometric data association & worker permissions
|   |   |   |— logs/      # Historical audit trail & audit generation
|   |   |— lib/         # DB clients & utility wrappers
|   |   |— types/       # Shared TypeScript definitions
|   |— firmware/       # Hardware Implementation (C++)
|   |   |— src/        # ESP32 Source Code
|   |   |   |— gateway.cpp # Biometric handling & Server communication
|   |   |   |— field_controller.cpp # Power toggling & Voltage monitoring
|   |   |   |— esp_now_manager.cpp # Fast low-latency node-to-node logic
|   |   |   |— failsafe.cpp  # Shunt trip & TDR logic
|   |   |— lib/        # Hardware libraries (Adafruit_Fingerprint, etc.)
|   |   |— platformio.ini # PlatformIO build configuration
|   |— docs/          # Design & CAD Documentation
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5. System Architecture

The system is divided into three primary operational nodes to ensure redundancy and safety:

1. **The Gateway (ESP32 #1):** Handles the **AS608 Biometric Scanner**. Upon a successful scan, it broadcasts a high-speed command via **ESP-NOW** to the Field Controller and simultaneously logs the event to the **PostgreSQL** database via the Next.js API.
2. **The Field Controller (ESP32 #2):** Receives commands to toggle the control relays and physically cut power to the machinery.
3. **The Feedback Loop & Failsafe:** The **ZMPT101B Voltage Sensor** continuously monitors the load side. If the system commands a shutdown but power remains (e.g., due to a welded relay), the **Time Delay Relay** triggers the **Shunt Trip** to mechanically trip the **Main Circuit Breaker**.

6. System Workflow (IPO Model)

Input

- **Hardware:** Dual ESP32, AS608 Scanner, ZMPT101B Sensor, Shunt Trip Breaker, and TDR .
- **Software:** Next.js Dashboard, Drizzle ORM, PostgreSQL, and PlatformIO .
- **Knowledge:** LOTO protocols, biometric operations, and system integration.

Process

- **Designing:** 2D/3D modeling (AutoCAD/SketchUp) and physical assembly of the MCB/Pull Box structure .
- **Programming:** Encoding biometric algorithms, ESP-NOW communication logic, and automated failsafe responses .
- **Testing:** Simulating fingerprint scans, measuring isolation response times, and verifying failsafe reliability during simulated relay failures.

Output

- **E-LOCK:** A fully functional prototype ensuring secure, accountable, and automated electrical isolation through multi-layered biometric and mechanical verification.

7. Significance of the Study

- **Workforce Protection:** Prevents accidental start-ups by verifying the physical presence of authorized workers via non-transferable biometric data.
- **Safety Compliance:** Replaces error-prone manual logs with a real-time, automated dataset, assisting safety officers in workplace audits and regulatory compliance .
- **Future Innovation:** Establishes a foundational blueprint for **Industrial Internet of Things (IIoT)** safety infrastructures.